



# ROBERT H. SMITH SCHOOL OF BUSINESS

**DATA-DRIVEN STRATEGIES TO IMPROVE  
RELIABILITY, SAFETY AND RESOURCE ALLOCATION  
FOR WASHINGTON METROPOLITAN AREA TRANSIT  
AUTHORITY (WMATA)**

**DATABASE MANAGEMENT SYSTEMS**

**BUDT703\_Project\_0506\_09\_Design**

**MCC SOLUTIONS**

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# 1. Project Introduction and Mission

## 1.1 The WMATA Business Problem

While WMATA operates a large and complex transit system, its operational data, including ridership volumes, service reliability metrics, and safety incidents, exist in isolated systems that lack analytical integration. These disconnected datasets make it challenging to detect performance trends, identify root causes of service issues, or correlate reliability and safety factors with customer ridership behaviors.

### 1.1.1 Mission Statement

To provide Washington Metropolitan Area Transit Authority (WMATA) with an in-depth analysis that breaks down historical performance, ridership, and safety data. Our goal is to assist with data-driven decision-making to improve service reliability, enhance customer safety, and optimize resource allocation across the Metrorail and Metrobus systems.

## 1.2 Mission Objectives

1. **Analyze Ridership Trends:** Track and compare data for the period 2022-2024 relating to ridership patterns for each Metrorail station and Metrobus route, identifying high-growth and low-performing areas.
2. **Correlate service performance with reliability:** Evaluate operational metrics such as on-time departures to determine how reliability affects customer experience and ridership.
3. **Enhance safety and security insights:** Identify and map safety incidents by Metrorail station and line.
4. **Support Performance Ranking and Resource Allocation:** Enable ranking of stations and routes by ridership, reliability, and safety metrics to inform planning, maintenance prioritization, and service redesign initiatives.

## 1.3 Group/Brand Name



**Brand Name: MCC SOLUTIONS**

MCC Solutions takes its name from the initials of the founding team members. This naming strategy reflects the collaborative nature of the group and highlights how the firm's analytical capabilities arise from pooling individual expertise. As a professional brand, the name represents the team's cohesion and commitment to providing clients with thorough, evidence-based analysis.

## 2. Business Terms, Facts, Attributes, and Identifiers

Understanding business terms, facts, attributes, and identifiers is critical for creating an accurate information model for the WMATA Performance analysis. This section formally defines these concepts within the context of the Entity-Relationship Diagram (ERD) developed for this project. The documentation ensures the conceptual model faithfully captures how the organization defines and manages ridership, service performance, and safety information.

### 2.1 Business Terms

- **WMATA:** Washington Metropolitan Area Transit Authority, the client.
- **Bus Route:** A Metrobus service line with scheduled trips and performance metrics.
- **Station:** A physical Metrorail location where entries, exits, and safety events are recorded.
- **Service Type:** The type of transit service (e.g., Metrorail, Metrobus, MetroAccess).
- **On-Time Performance (OTP):** A key metric measuring service reliability. For buses, it's typically 'on-time' if the bus departs between 2 minutes early and 7 minutes late.
- **Ridership:** The count of passenger trips. For Metrorail, this is often measured by station entries/exits. For Metrobus, it's boardings.
- **Part I Crime:** Serious crimes as defined by the FBI, used as a key security metric.
- **Calendar Date:** A specific operational day with attributes such as month, quarter, year, and weekday/weekend classification.
- **Rail Performance:** metrics for each station, including ridership, crime incidents, injuries, and escalator availability.
- **Bus Performance:** Metrics for each bus route, including boardings and on-time performance.
- **Reliability:** A measure of operational consistency (e.g., on-time performance, completed vs. scheduled trips).
- **Safety:** Incidents affecting customers, such as injuries and reported crimes.

### 2.2 Business Facts

- Each Bus Route records BusPerformance metrics for multiple Calendar Dates over time.
- Each Station records RailPerformance metrics for multiple Calendar Dates over time.
- Ridership is measured and recorded daily as entries/exits (rail) or boardings (bus).
- Service reliability is evaluated daily for each station and route.
- Safety incidents and customer injuries are tracked per station per day.
- Performance metrics enable temporal analysis across days, months, quarters, and years.
- Each Calendar Date can have performance records for multiple stations and routes.

### 2.3 Attributes and Classification

### 2.3.1 Station Attributes

- stationID
- stationName
- stationPrimaryLine
- stationSecondaryLine
- stationAddress
- stationJurisdiction

### 2.3.2 BusRoute Attributes

- busRouteID
- busRouteName
- busRouteServiceType

### CalendarDate Attributes

- calendarDateID
- calendarDateFull
- monthName
- monthNumber
- quarter
- year
- dayOfWeek

### 2.3.3 RailPerformance Attributes

- stationID (FK)
- calendarDateID (FK)
- railRidershipEntries
- railRidershipExits
- railCrimeIncidentCount
- railCustomerInjuryCount
- railEscalatorAvailabilityPct
- =totalRidership (derived = entries + exits)

### 2.3.4 BusPerformance Attributes

- busRouteID
- calendarDateID
- busBoardings
- busOnTimeDepartures
- busTotalDepartures
- busTotalScheduledTrips
- busActualTrips

## 2.3 Identifiers

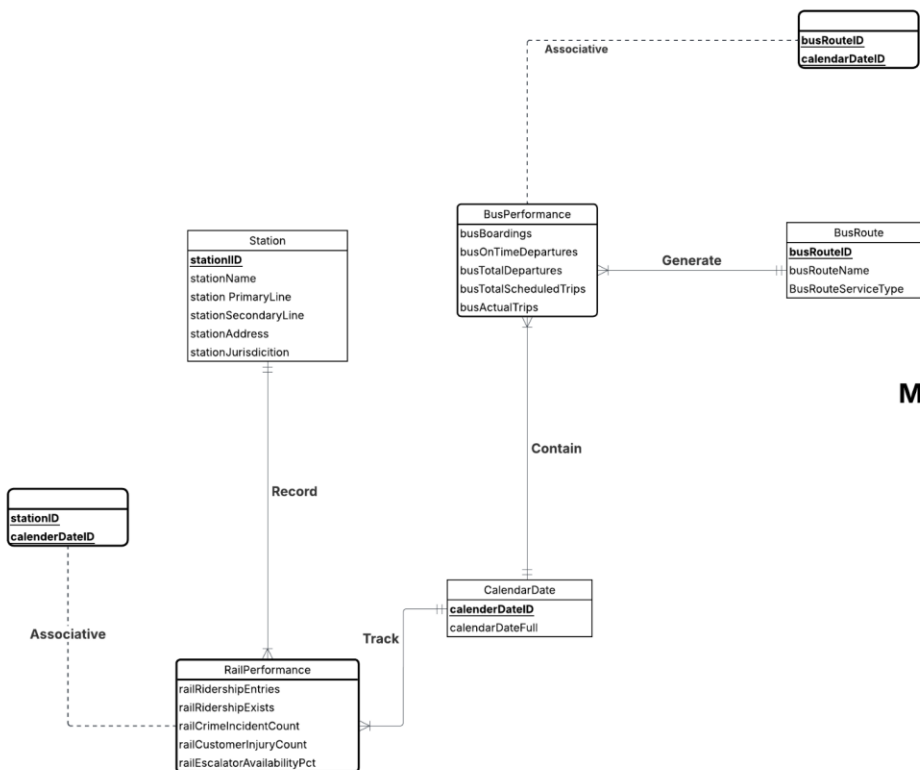
### Strong Identifiers (Primary Keys)

- Station: **stationID**
- BusRoute: **busRouteID**
- CalendarDate: **calendarDateID**

### Composite Identifiers for Associative Entities

- RailPerformance: (**stationID**, **calendarDateID**)
- BusPerformance: (**busRouteID**, **calendarDateID**)

## 3. Conceptual Database Design (Entity Relationship Diagram)



**Metro Performance . by  
MCC Solutions**

The conceptual Entity-Relationship Diagram (ERD) captures the core business entities and their associations. It conforms to project naming rules:

- Entities and relationships use PascalCase (e.g., RailPerformance).

- Attributes use Hungarian-style lowerCamelCase ( e.g railRidershipEntries).
- Primary keys are underlined; foreign keys are omitted in conceptual ERD per project guidelines.
- Crow's foot notation indicates cardinalities.

## 4. Logical Database Design (Relational Schema)

### 4.1 Relational Schema

- Station (stationID, stationName, stationPrimaryLine, stationSecondaryLine, stationAddress, stationJurisdiction)
- BusRoute (busRouteID, busRouteName, busServiceType)
- CalendarDate (calendarDateID, calendarFullDate)
- RailPerformance (stationID, calendarDateID, railRidershipEntries, railRidershipExits, railCrimeIncidentCount, railCustomerInjuryCount, railEscalatorAvailabilityPct)
- BusPerformance (busRouteID, dateID, busBoardings, busOnTimeDepartures, busTotalDepartures, busTotalScheduledTrips, busActualTrips)

### 4.2 Business Rules and Referential Integrity

#### 4.2.1 Defining Business Rules with Cardinality Constraints:

- **Station to RailPerformance (1:N):** One Station can have many RailPerformance records (one for each day).
- **CalendarDate to RailPerformance (1:N):** One CalendarDate can have many RailPerformance records (one for each station).
- **BusRoute to BusPerformance (1:N):** One BusRoute can have many BusPerformance records (one for each day).
- **CalendarDate to BusPerformance (1:N):** One CalendarDate can have many BusPerformance records (one for each bus route).
- **Station and CalendarDate:** A station has many days of performance data, and a day has performance data for many stations. The RailPerformance table resolves this M:N relationship, with each row representing the performance of one station on one specific day. **Composite PK must be unique eg stationID, calendarDateID**
- **BusRoute and CalendarDate:** A bus route has many days of performance data, and a day has performance data for many bus routes. The BusPerformance table resolves this M:N relationship. **Composite PK must be unique eg busRouteID, calendarDateID.**

### 4.3 Business Rules and Referential Integrity Actions

- **[R1]** When a Station's stationID is updated, all corresponding performance records in RailPerformance must be updated to maintain the link.
- **[R2]** A Station record cannot be deleted if it is referenced by any records in the RailPerformance table. This prevents the "orphaning" of historical performance data.
- **[R3]** When a CalendarDate's dateID is updated, all corresponding records in RailPerformance and BusPerformance must be updated to reflect the change.
- **[R4]** A CalendarDate record cannot be deleted if it is referenced by any records in RailPerformance or BusPerformance, as this would corrupt the historical timeline.
- **[R5]** When a BusRoute's busRouteID is updated, all corresponding performance records in BusPerformance must be updated to maintain the link.
- **[R6]** A BusRoute record cannot be deleted if it is referenced by any records in the BusPerformance table, preventing the loss of historical route data.

| Relation        | Foreign Key | Base Relation | Primary Key     | Business Rule | ON DELETE | Business Rule | ON UPDATE |
|-----------------|-------------|---------------|-----------------|---------------|-----------|---------------|-----------|
| RailPerformance | stationID   | Station       | stationID       | R2            | NO ACTION | R1            | CASCADE   |
| RailPerformance | dateID      | Calendar Date | calendar DateID | R4            | NO ACTION | R3            | CASCADE   |
| BusPerformance  | busRouteID  | BusRoute      | busRoute ID     | R6            | NO ACTION | R5            | CASCADE   |
| BusPerformance  | dateID      | Calendar Date | calendar DateID | R4            | NO ACTION | R3            | CASCADE   |

## 4.4 Sample Data

- **Station**
  - stationID: A01
  - stationName: "College Park-U of Md"
  - stationPrimaryLine: "Green"
  - stationSecondaryLine: "Yellow"
  - stationAddress: "4901 Calvert Rd, College Park, MD 20740"
  - stationJurisdiction: "Prince George's County"
- **BusRoute**
  - busRouteID: 10A
  - busRouteName: "10A-Fort Totten"
  - busRouteServiceType: "Local"
- **CalendarDate**

- dateID: "2024-10-27"
  - fullDate: "2024-10-27"
  - monthName: "October"
  - quarter: 4
  - year: 2024
  - dayOfWeek: "Monday"
- **RailPerformanceFac**
    - calenderDateID: "2022-01-01"
    - stationID: A01
    - railRidershipEntries: 287
    - railRidershipExits: 307
    - railCrimeIncidentCount: 0.001751
    - railCustomerInjuryCount: 0.000092
    - railEscalatorAvailabilityPct: 98.5
- **BusPerformance**
    - calendarDateID: "2023-01-04"
    - busRouteID: 10A
    - busPerformanceBoardings: 1689
    - busPerformanceTimeDepartures: 62
    - busPerformanceTotalDepartures: 82
    - busPerformanceScheduledTrips: 84
    - busPerformanceActualTrips: 82

## Conclusion

This Design Document provides the formal specification linking WMATA's analytical requirements to a comprehensive database architecture. It articulates the domain vocabulary, entity structures, relational mappings, business rules, and integrity constraints necessary for SQL schema construction and query development. The document constitutes the technical foundation upon which all subsequent project components, including database implementation, analytical workflows, and business intelligence dashboards, will be built.